An algebraic approach to manifolds and the Serre-Swan Theorem

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Contents

1 Introduction **1**

2 An algebraic approach to manifolds **2**
2.1 Preliminaries 2
2.2 Manifolds as spectra of $\mathbb{R}$-algebras 3
2.3 Differential calculus 7

3 Vector bundles **8**
3.1 Bundles and their properties 8
3.2 Serre-Swan Theorem 10

4 Algebraic geometry **13**
4.1 Correspondence between vector bundles and locally free sheaves 13
4.2 Serre's Theorem 14

Bibliography **15**

Abstract

In this note, we aim to give an alternative definition for smooth manifolds that involves studying the maximal ideal spectra of commutative algebras over a field. We try to see analogues of well-known topological concepts in this algebraic setting, show their equivalence with the usual definitions, and as an application, prove the Serre-Swan Theorem which relates the sections of vector bundles over a connected manifold to finitely generated projective modules over its ring of smooth functions. Finally, we move to the realm of algebraic geometry to see an analogue of this theorem for vector bundles over schemes.

1 Introduction

The motivation for an algebraic approach to studying smooth manifolds originates, unsurprisingly to an extent, from the developments in algebraic geometry. By the early 20th century, the limitations of working with affine algebraic varieties, and the reliance of the theory on algebraically closed fields, necessitated the need for the study of more general objects. Mimicking the model for smooth manifolds, the concept of an abstract variety emerged, which was obtained by gluing together affine varieties. Further extension towards the mid 20th century laid the foundations of schemes, led by J.P. Serre and C. Chevalley and eventually formalised by Grothendieck. The sheaf theoretic

approach for schemes allowed one to use tools similar to the ones in the theory of manifolds, and gave way to a plethora of techniques for dealing with problems in algebraic geometry.

The concept of the spectrum of a ring, that thus originated, was something quite general which can be used to recover the original theory of smooth manifolds. Studying the maximal spectra of certain special $\mathbb{R}$-algebras gave a coordinate free approach for defining smooth manifolds, that makes certain definitions simpler and much more natural. The strength of this approach becomes apparent when we build the theory of differential calculus as a part of commutative algebra, and specialize it to study differential forms, jet bundles and vector fields over a manifold. While we develop some of this theory in this report, the theory of Kähler differentials is quite extensive in itself and will not be pursued here. A more detailed account can be found in [1] or [5].

While this approach may have the occasional disadvantage of obfuscating the geometrical and intuitive aspects of the theory of manifolds, the equivalence of the geometrical and the algebraic approaches allows each to supplement the other, making the study of manifolds much more convenient.

2 An algebraic approach to manifolds

2.1 Preliminaries

Here we collect some results from basic calculus and point set topology that will be useful for us throughout. Proofs for all of these can be found in [1].

Lemma 2.1. *For any open set $U \subset \mathbb{R}^n$, there exists $f \in C^\infty(\mathbb{R}^n)$ such that $f(x) > 0$ for all $x \in U$, and $f(x) = 0$ otherwise.*

Proposition 2.2. *Given a non-empty open subset U of $\mathbb{R}^n$, there exists a smooth function f over U, such that for all $\lambda \in \mathbb{R}$, $f^{-1}(\lambda)$ is a compact subset of $\mathbb{R}^n$, i.e., all the level surfaces of f are compact.*

Both of the above results can be proven for arbitrary manifolds, in fact, by using a partitions of unity argument.

A domain $U \in \mathbb{R}^n$ is said to be starlike with respect to $z \in U$ if for all $x \in U$, the segment connecting z and x lies entirely in U. Next, we define for a multi-index $\mu = (i_1, i_2, \ldots, i_n)$ the following notation:

$$|\mu| = \sum_{j=1}^{n} i_j$$

$$\frac{\partial^{|\mu|}}{\partial x^\mu} = \frac{\partial^{|\mu|}}{\partial x_1^{i_1} \partial x_2^{i_2} \ldots \partial x_n^{i_n}}$$

$$(x-a)^\mu = \prod_{j=1}^{n} (x_j - a_j)^{i_j}$$

$$\mu! = i_1! i_2! \ldots i_n!$$

Lemma 2.3 (Hadamard)**.** *Given a smooth function f on a domain U that is starlike with respect to some z, we can express f as:*

$$f(x) = \sum_{|\mu|=0}^{m} \frac{(x-z)^\mu}{\mu!} \frac{\partial^{|\mu|} f}{\partial x^\mu}(z) + \sum_{|\mu|=m+1} (x-z)^\mu g_\mu(x),$$

for some $g_\mu \in C^\infty(U)$.

2.2 Manifolds as spectra of $\mathbb{R}$-algebras

In this section, we develop the theory of manifolds as the maximal spectra of *geometric* and *smooth* $\mathbb{R}$-algebras. Most of the theory goes through if we work with an arbitrary field K of characteristic 0 instead of $\mathbb{R}$. Throughout the report, it will be assumed that $\mathcal{F}$ is a commutative and associative $\mathbb{R}$-algebra with unity, and that all algebra homomorphisms take unity to unity. All tensor products will be taken with respect to $\mathbb{R}$, unless stated otherwise. The chief example to have in mind is $\mathcal{F} = C^\infty(M)$ for some smooth manifold M.

Given an $\mathbb{R}$-algebra $\mathcal{F}$, we define the set of $\mathbb{R}$-points of $\mathcal{F}$ as the set $|\mathcal{F}|$ of all $\mathbb{R}$-algebra homomorphisms from $\mathcal{F}$ to $\mathbb{R}$. Then, any $f \in \mathcal{F}$ naturally defines a function over $|\mathcal{F}|$ as follows:

$$\tilde{f} : |\mathcal{F}| \to \mathbb{R}$$

$$x \mapsto x(f).$$

The set of functions $\tilde{f}$ for $f \in \mathcal{F}$ forms an $\mathbb{R}$-vector space with respect to addition of functions and pointwise scalar multiplication. The set $|\mathcal{F}|$ will be our candidate for being defined as a smooth manifold, once we impose suitable conditions on $\mathcal{F}$. The first condition we wish to enforce is that for all $f \in \mathcal{F}$, the function $\tilde{f}$ should be distinct, that is, we want $f \neq g \implies \tilde{f} \neq \tilde{g}$. This is equivalent to having the following condition:

Definition 2.1. $\mathcal{F}$ is called a geometric $\mathbb{R}$-algebra if

$$\bigcap_{x \in |\mathcal{F}|} \ker(x) = 0.$$

Henceforth, we will identify the function $\tilde{f}$ with f itself. Next, we define the topology on $|\mathcal{F}|$ to be the weakest topology that that makes all the functions in $\mathcal{F}$ continuous. It's an easy observation that this induced topology is Hausdorff. The described association of a topological space to an $\mathbb{R}$-algebra is functorial:

Lemma 2.4. *Given an $\mathbb{R}$-algebra homomorphism $\phi : \mathcal{F}_1 \to \mathcal{F}_2$, we get an induced map on the topological spaces in the opposite direction $|\phi| : |\mathcal{F}_2| \to |\mathcal{F}_1|$ that is continuous.*

Next, in order to capture the quality of *smoothness*, we need some sort of local properties on our algebras, motivating the following definition.

Definition 2.2. Given a geometric $\mathbb{R}$-algebra $\mathcal{F}$, for any subset $V \subset |\mathcal{F}|$, we define $\mathcal{F}|_V$ to be the set of functions $f : V \to \mathbb{R}$ such that for all $x \in V$, there exists an open neighbourhood U_x of x in V such that the restriction of f to U_x equals the restriction to U_x of some $g \in \mathcal{F}$.

As the elements of $\mathcal{F}$, when restricted to any subset V of $|\mathcal{F}|$, satisfy the above properties, we get a map $\rho : \mathcal{F} \to \mathcal{F}|_V$, which is obviously injective when $V = |\mathcal{F}|$. However, the map may not necessarily be surjective as is clear from the following example.

Example 2.1. Let $\mathcal{P}$ denote the $\mathbb{R}$-algebra of polynomials with real coefficients in one variable. Then, the space $|\mathcal{P}|$ is $\mathbb{R}$ and it is easy to see that any function on $\mathbb{R}$ that locally looks like a polynomial, is a polynomial itself and so, the map ρ is surjective. On the other hand, let $\mathcal{B}$ denote the $\mathbb{R}$-algebra of continuous and bounded real-valued functions on $\mathbb{R}$. The space $|\mathcal{B}|$ is $\mathbb{R}$ again, but this time, the map ρ is definitely not surjective as $\mathcal{B}|_{\mathbb{R}}$ is the space of all continuous functions on $\mathbb{R}$, of which, $\mathcal{B}$ is a proper subspace.

Definition 2.3. A geometric $\mathbb{R}$-algebra is said to be complete if the map $\rho : \mathcal{F} \to \mathcal{F}|_{|\mathcal{F}|}$ is an isomorphism, i.e., if any function on $|\mathcal{F}|$ that locally likes like an element of $\mathcal{F}$ must actually be an element of $\mathcal{F}$.

While the above tools will, almost always, be sufficient for our study of smooth manifolds, later, when we talk about vector bundles, it would be useful to have the concept of a C^∞-closed $\mathbb{R}$-algebra.

Definition 2.4. A geometric $\mathbb{R}$-algebra $\mathcal{F}$ is said to be C^∞-closed if for any collection of elements $f_1, f_2, \ldots, f_n \in \mathcal{F}$ and any function $f \in C^\infty(\mathbb{R}^n)$, there is some element $g \in \mathcal{F}$ such that for all $a \in |\mathcal{F}|$, $g(a) = f(f_1(a), f_2(a), \ldots, f_n(a))$.

This basically means that the algebra $\mathcal{F}$ is, in some sense, closed under smooth operations. Given an arbitrary geometric $\mathbb{R}$-algebra $\mathcal{F}$, we can construct a C^∞-closed $\mathbb{R}$-algebra $\overline{\mathcal{F}}$ and an $\mathbb{R}$-algebra map $i : \mathcal{F} \to \overline{\mathcal{F}}$, such that we have the universal property that any map from $\mathcal{F}$ to a C^∞-closed $\mathbb{R}$-algebra factors through $\overline{\mathcal{F}}$. The algebra $\overline{\mathcal{F}}$ is called the smooth envelope of $\mathcal{F}$, and is constructed by a standard category theoretical argument. It is obvious that the $\mathbb{R}$-algebra $C^\infty(\mathbb{R}^n)$ is itself C^∞-closed, which would imply that all smooth algebras are C^∞-closed (See next definition). With this, we have the required machinery to give the algebraic definition of a smooth manifold.

Definition 2.5. A complete, geometric $\mathbb{R}$-algebra $\mathcal{F}$ is called smooth if there is a countable covering $\{\mathcal{U}_k\}$ of the space $|\mathcal{F}|$ such that for all k, the algebra $\mathcal{F}|_{\mathcal{U}_k}$ is isomorphic to $C^\infty(\mathbb{R}^n)$. Given such an algebra $\mathcal{F}$, we define $|\mathcal{F}|$ to be a smooth manifold and refer to $\mathcal{F}$ as the algebra of smooth functions over this manifold.

The integer n is called the dimension of the manifold M. A suitable modification of the above definition (replacing $\mathbb{R}^n$ by a half space $\mathbb{H}^n$) can be used for defining manifolds with boundaries, but we will not pursue that here. Right now, the above definition may seem incongruous with the usual definition of a smooth manifold. This incompatibility will be cleared shortly when we prove the equivalence between the above definition and the standard definition of a manifold.

Example 2.2. 1. Consider the algebra $\mathcal{F}$ of smooth functions on $\mathbb{R}$ such that for all $f \in \mathcal{F}$, $f(x+1) = f(x)$ for all $x \in \mathbb{R}$. Then, by identifying $\mathcal{F}$ as the algebra of smooth real valued functions on the circle $\mathbb{S}^1$, it becomes clear that $|\mathcal{F}| = \mathbb{S}^1$. We claim that $\mathcal{F}$ is a smooth algebra. That it is geometric and complete is clear. Next, covering $\mathbb{R}$ by the open sets $U_1 = \mathbb{R} \setminus \mathbb{Z}$ and $U_2 = \mathbb{R} \setminus (\mathbb{Z} + \frac{1}{2})$, we can see that their images in $\mathbb{S}^1$ correspond to $\mathbb{S}^1 \setminus \{\text{some pt}\}$. Now, removing a point from $\mathbb{S}_1$ makes it diffeomorphic to $\mathbb{R}$, and so, $\mathcal{F}|_{U_i} \cong C^\infty(\mathbb{R})$. As $U_1 \cup U_2 = \mathbb{R}$, this proves that $\mathcal{F}$ is smooth.

2. By a similar argument as above, if we consider the algebra:

$$\mathcal{F} := \{f \in C^\infty(\mathbb{R}^n) : f(x+c) = f(x) \text{ for all } x \in \mathbb{R}^n, c \in \mathbb{Z}^n\},$$

we get a smooth algebra that corresponds to the manifold $|\mathcal{F}| = \mathbb{T}^n$.

3. The particular case above when $n = 2$ corresponds to $\mathbb{S}^1 \times \mathbb{S}^1 = \mathbb{T}^2$, the 2-dimensional torus. Tweaking the algebra $\mathcal{F}$ slightly, if we look at:

$$\mathcal{F} := \{f \in C^\infty(\mathbb{R}^2) : f(x,y) = f(x,y+1) = f(x+1,1-y) \text{ for all } x,y \in \mathbb{R}\},$$

we get the Klein bottle K. The argument for $\mathbb{S}_1$ can be repeated to get a covering of $|\mathcal{F}|$ by 4 open sets where the restriction of $\mathcal{F}$ is isomorphic to $C^\infty(\mathbb{R}^2)$, which would show that K is a smooth manifold.

Definition 2.6. Given a manifold M with the algebra of smooth functions $\mathcal{F}$, a subset $N \subset M$ is called a smooth submanifold if the algebra $\mathcal{F}|_N$ is a smooth $\mathbb{R}$-algebra. In such a case, we refer to $\mathcal{F}|_N$ as the algebra of smooth functions over N.

Remark 2.1. There is a subtlety related to the above definition. Considering the $\mathbb{R}$-algebra $\mathcal{F}|_N$, we can also talk about the manifold $|\mathcal{F}|_N|$, and it isn't a priori clear that this manifold is N itself. This is a matter of routine bookkeeping and definition checking, and is easily verified to be actually true.

Given manifolds M_1 and M_2, it is natural to consider the product topological space $M_1 \times M_2$ and talk about the algebra of smooth functions over $M_1 \times M_2$. One obvious candidate for this is $\mathcal{F}_1 \otimes \mathcal{F}_2$, where $\mathcal{F}_1$ and $\mathcal{F}_2$ are, respectively, the algebras of smooth functions over M_1 and M_2. It turns out that this guess is almost right:

Proposition 2.5. *Given geometric $\mathbb{R}$-algebras $\mathcal{F}_1$ and $\mathcal{F}_2$, the algebra $\mathcal{F}_1 \otimes \mathcal{F}_2$ is geometric too, and we have that $\overline{\mathcal{F}_1 \otimes \mathcal{F}_2}$ is the algebra of smooth functions over the manifold $|\mathcal{F}_1| \times |\mathcal{F}_2|$.*

Example 2.3.
1. By the universal property of tensor products, we have that $C^\infty(\mathbb{R}^{m+n}) \cong \overline{C^\infty(\mathbb{R}^m) \otimes C^\infty(\mathbb{R}^n)}$, and so, we have (as we'd expect) that the Cartesian product of the manifolds $\mathbb{R}^m$ and $\mathbb{R}^n$ is $\mathbb{R}^{m+n}$.

2. Slightly less obvious, but equally easy to check, is the assertion that the manifold $\mathbb{T}^n$ is the Cartesian product of $\mathbb{T}^{n-1}$ and $\mathbb{S}_1$, and thus, the same as $\underbrace{\mathbb{S}^1 \times \mathbb{S}^1 \cdots \times \mathbb{S}^1}_{n \text{ times}}$.

Definition 2.7. Given smooth manifolds M_1 and M_2 with $\mathcal{F}_1$ and $\mathcal{F}_2$ as their respective algebras of smooth functions, a map $f : M_1 \to M_2$ is called a smooth map if $f = |\phi|$, where $\phi : \mathcal{F}_2 \to \mathcal{F}_1$ is an $\mathbb{R}$-algebra homomorphism. A smooth map that is a bijection, such that the inverse map is also smooth, is called a diffeomorphism.

Now, we are ready to prove the main theorems of this section that establish an equivalence between the two apparently disparate definitions of a smooth manifold.

Theorem 2.6. *Given a manifold M in the usual sense, the algebra $\mathcal{F} = C^\infty(M)$ is a smooth $\mathbb{R}$-algebra, and the map:*

$$\phi : M \to |\mathcal{F}|$$
$$x \mapsto ev_x$$

(where ev_x denotes the 'evaluation by x' map) is a homeomorphism.

Proof. Let A be an atlas on M. Then, for any $x \in M$, there exists a neighbourhood $U \subset M$ of x that is homeomorphic to $\mathbb{R}^n$. Now, as smoothness is local, $\mathcal{F}|_U$ is the algebra of smooth real valued functions on U, and is thus equal to $C^\infty(U)$ which is isomorphic to $C^\infty(\mathbb{R}^n)$ as an $\mathbb{R}$-algebra. As we can cover M by such open sets, we get that $\mathcal{F}$ is a smooth algebra. That the above map is continuous, and has a continuous inverse, is almost tautological and is just a matter of unwrapping the definitions.

Injectivity is clear using the fact that M is Hausdorff, and thus, smooth functions separate points in M. In order to prove surjectivity, suppose $\theta \in |\mathcal{F}|$ such that $\theta \notin \mathrm{Im}(\phi)$. Pick $f \in C^\infty(M)$ all of whose level surfaces are compact subsets of M, and let $\lambda = \theta(f)$. As $\theta \neq ev_x$ for any $x \in M$, we have a function $f_x \in C^\infty(M)$ for each x such that $f_x(x) \neq \theta(f_x)$. Consider the opens sets U_x:

$$U_x := \{y \in M : f_x(y) \neq \theta(f_x)\},$$

which cover M. Taking $N = f^{-1}(\lambda)$ which is compact, the above sets give a covering for N, and so, we must have a finite subcover by, say, $U_{x_1}, U_{x_2}, \ldots, U_{x_m}$. Define the function:

$$h = (f - \lambda)^2 + \sum_{i=1}^{m} (f_{x_i} - \theta(f_{x_i}))^2.$$

Then, $\theta(h) = 0$, whereas, h is a non-vanishing function on M. Thus, h is a unit in $\mathcal{F}$, and hence, so must be $\theta(h)$. The last two statements contradict each other, and so, the proof is complete. □

Theorem 2.7. *If $\mathcal{F}$ is a smooth $\mathbb{R}$-algebra, $|\mathcal{F}|$ is a manifold in the usual sense, that is, there exists an atlas A on $|\mathcal{F}|$ such that the map*

$$\phi : \mathcal{F} \to C^\infty(M)$$

$$f \mapsto ev_f$$

(where ev_f denotes the 'evaluation by f' map) is an algebra isomorphism, where $M := |\mathcal{F}|$.

Proof. As $\mathcal{F}$ is smooth, we can cover M by open sets U such that $\mathcal{F}|_U \cong C^\infty(\mathbb{R}^n)$. We will use these open sets to define an atlas on M. By the local nature of smoothness, we have that the set U is homeomorphic to $|\mathcal{F}|_U|$. Next, the algebra homomorphism between $\mathcal{F}|_U$ and $C^\infty(\mathbb{R}^n)$ gives a homeomorphism between $|\mathcal{F}|_U|$ and $\mathbb{R}^n$. This gives us a homeomorphism between U and $\mathbb{R}^n$. Thus, we get an atlas on M, where the transition maps are smooth as the maps on the corresponding algebras of smooth functions are $\mathbb{R}$-algebra homomorphisms.

Next, we want to show that the map ϕ defined in the statement of theorem is well-defined and is an isomorphism. It is well-defined because any $f \in \mathcal{F}$ defines a smooth function on M with respect to the topology defined above. Injectivity of ϕ follows directly from the fact that $\mathcal{F}$ is geometric. For surjectivity, we note that given any smooth function f on $|M|$, we can consider its restrictions to the the open sets U as above, and then, we can glue together the restrictions in $\mathcal{F}|_U$ (due to the completeness of $\mathcal{F}$) to get an element of $\mathcal{F}$ that maps to f under ϕ. □

Remark 2.2. The two theorems above define an equivalence of categories between the usual category of smooth manifolds with smooth functions as morphisms, and the category defined using Definitions 2.5 and 2.7.

2.3 Differential calculus

In this subsection, we develop techniques that allow us to describe tangents to manifolds in terms of the corresponding $\mathbb{R}$-algebras. Most of the theory here works when $\mathbb{R}$-algebras arbitrary commutative rings, but we will not be working with that degree of generality.

Definition 2.8. A tangent vector to a manifold $M = |\mathcal{F}|$ at a point $x \in M$ is a map $v : C^\infty(M) \to \mathbb{R}$ which satisfies:

- $v(\sum_{i=1}^{n} \lambda_i f_i) = \sum_{i=1}^{n} \lambda_i v(f_i)$ for all $\lambda_i \in \mathbb{R}$ and $f_i \in C^\infty(M)$.
- $v(fg) = g(x)v(f) + f(x)v(g)$ for all $f, g \in C^\infty(M)$.

Such a definition does not come as a surprise as we are used to interpreting tangent vectors as derivations over the algebra of smooth functions. The advantage that this definition proffers is that defining an $\mathbb{R}$-vector space structure to the space of all tangent vectors at a point is completely natural, when we define the sum of 2 tangent vectors as the sum of the respective functions, and similarly for scalar multiplication. This space is known as T_xM. And as one would expect, the equivalence of this with the usual definition is not very hard:

Proposition 2.8. *For $x \in M$, having fixed local coordinates $(x_1, x_2, \ldots, x_n)$ in a neighbourhood U of x in M, any $v \in T_xM$ is expressible as*

$$v = \sum_{i=1}^{n} \alpha_i \frac{\partial}{\partial x_i},$$

where $\alpha_i \in \mathbb{R}$, and $\frac{\partial}{\partial x_i}$ denotes taking the partial derivative in the ith direction at the point x.

Given a smooth map between manifolds $|\phi| : M \to N$ that corresponds to an algebra homomorphism $\phi : \mathcal{F}_1 \to \mathcal{F}_2$, for all $x \in M$, we get a map $d\phi_x : T_x(M) \to T_{|\phi|(x)}(N)$ given by $v \to v \circ \phi$. This is called the differential of the map ϕ at the point x.

With the definition of tangent vectors and the derivative of a map at our disposal, the construction of tangent and cotangent bundles on a manifold can be done in the usual fashion. we will later see that the definition of vector bundles can be put in an algebraic framework, and our usual bundles do satisfy those conditions. Before moving on, we make a passing remark that is of independent interest. The fiber of the cotangent bundle at a point can be represented in a nice algebraic fashion, just as is the case in algebraic geometry. For a point $z \in M$, define

$$\mu_z = \{f \in C^\infty(M) : f(z) = 0\}.$$

Proposition 2.9. *T_z^*M is naturally isomorphic to the space μ_z/μ_z^2.*

Proof. We can express $C^\infty(M) = \mathbb{R} \oplus \mu_z$, where $\mathbb{R}$ identifies the space of constant functions on M. Now, we have the surjection:

$$d_z : C^\infty(M) \to T_z^*M$$
$$f \mapsto d_z(f).$$

Writing $C^\infty(M)$ as a direct sum as above, the constant functions evaluate to 0 under d_z. Also, by Lemma 2.3, any element of μ_z evaluates to zero under d_z if and only if it lies in μ_z^2. Thus, the quotient $\frac{\mu_z}{\mu_z^2}$ maps isomorphically onto the image, which was exactly the claim. □

Inspired by the above proposition, defining $J^1_z(M) := \frac{C^\infty(M)}{\mu_z^2}$ as fibers and mimicking the construction of the tangent bundle, we get the jet bundle of order 1 over M. For an arbitrary positive integer l, we can construct the jet bundle of order l over M that has fiber at z $\frac{C^\infty(M)}{\mu_z^{l+1}}$. These can be morally though of as *differential operators of order l* over M. As we will not need them further, we will not justify this claim, and rather, just mention that these bundles come into play when we try to explicitly describe the R-algebra whose spectrum is the cotangent bundle of the manifold M.

Now, we move on to define the concept of vector fields. In our usual definitions, we define a vector field by defining tangent vectors at each point on the manifold, that vary smoothly. Motivated by this, as a tangent vector, for us, at a point $x \in M$ is a derivation on $C^\infty(M)$, we want to define a bunch of derivations, one corresponding to each point on the manifold. This leads us to the following definition.

Definition 2.9. An $\mathbb{R}$-linear map $X : C^\infty(M) \to C^\infty(M)$ is called a smooth vector field on M if it satisfies the Liebniz rule, i.e. $X(fg) = fX(g) + gX(f)$ for all $f, g \in C^\infty(M)$.

Using the equivalence between the two definitions of tangent vectors, it follows that the above definition for vector fields is equivalent to the usual one. Analogously, we can define a smooth vector field on a submanifold N of M as an $\mathbb{R}$-linear map $X : C^\infty(M) \to C^\infty(M)$. More generally, for a smooth map $\phi : N \to M$, we define a vector field along the map ϕ to be an $\mathbb{R}$-linear map $X : C^\infty(M) \to C^\infty(N)$ such that $X(fg) = X(f)\phi^*(g) + \phi^*(f)X(g)$.

3 Vector bundles

3.1 Bundles and their properties

For any subset $U \subset M$, we define the subset $S_U \subset C^\infty(M)$ of those smooth functions over M that do not vanish at any point in U. Then, S_U forms a multiplicative subset of the ring $C^\infty(M)$, and so, we can talk about the localised ring $S_U^{-1}(C^\infty(M))$. When $U = \{x\}$ is a singleton, this ring represents the ring of *germs* of smooth functions at the point x. Whereas when U is an open set, we have:

Lemma 3.1. *The $\mathbb{R}$-algebra $C^\infty(U)$ of smooth functions on U is canonically isomorphic to the algebra $S_U^{-1}(C^\infty(M))$.*

The above identification motivates the following definition of a fiber bundle in the algebraic setting.

Definition 3.1. Given smooth $\mathbb{R}$-algebras A, B, F, an injective homomorphism $i : A \hookrightarrow B$ is called a fiber bundle $|B|$ over $|A|$ with fiber $|F|$, if for all $x \in |A|$, there exists an open neighbourhood U of x in $|A|$ such that we have an algebra isomorphism $p : S_U^{-1}B \to \overline{S_U^{-1}A \otimes F}$ which makes the following diagram commute (where j denotes the natural inclusion into the tensor product):

$$\begin{array}{ccc} S_U^{-1}A & \xrightarrow{S_U^{-1}i} & S_U^{-1}B \\ \downarrow{\scriptstyle j} & \swarrow{\scriptstyle p} & \\ \overline{S_U^{-1}A \otimes F} & & \end{array} .$$

A bundle is called trivial if the above open set U can be chosen uniformly for every point in $|A|$, i.e., we can choose $U = |A|$. A section of the bundle defined above is a map in the reverse direction $\phi : B \to A$, such that $\phi \circ i = id$. The set of sections of a bundle π is denoted by $\Gamma(\pi)$. Given bundles $E_\pi \xrightarrow{\pi} M$ and $E_\mu \xrightarrow{\mu} M$, we say that π is a subbundle of μ if E_π is a submanifold of E_μ, and for every point in M, the fiber in E_π is a submanifold of its fiber in E_μ, such that the inclusion maps commute with the bundle maps.

Next, we define the Whitney sum, or the direct sum, of two bundles. Given injective $\mathbb{R}$-algebra maps $i_1 : A \hookrightarrow B$ and $i_2 : A \hookrightarrow C$, we define the map $i : A \to \overline{B \otimes_A C}$ to be their Whitney sum. This map is well-defined and injective by the universal property of tensor products. In category theoretic terms, this is a pullback or a fiber product (which we will also encounter in the algebraic geometry setting) and can be thought of, geometrically, as a bundle whose fibers are the Cartesian products of the fibers of the two bundles. We use the symbol $\oplus$ to denote Whitney sums.

Example 3.1.
1. The natural inclusion of the algebra of 1-periodic smooth functions over $\mathbb{R}$ into the algebra of smooth functions over $\mathbb{R}$ gives us that $\mathbb{R}$ is a bundle over $\mathbb{S}^1$ with fiber $\mathbb{Z}$, whose algebra of smooth functions corresponds to $\mathbb{Z}$-graded sequences of real numbers under pointwise addition and multiplication. More generally, we have that $\mathbb{R}^n$ is a fiber bundle over $\mathbb{T}^n$ with fiber $\mathbb{Z}^n$. Bundles with zero dimensional fibers are called coverings.

2. An open Möbius strip is a over $\mathbb{S}^1$ with fiber $\mathbb{R}$ such that its Whitney sum with itself is the solid torus, which is a trivial bundle over $\mathbb{S}^1$.

3. Lie groups are examples of manifolds whose tangent bundles are trivial. Such manifolds are called parallelizable.

Definition 3.2. A morphism (also called a bundle morphism) between bundles $M_\pi \xrightarrow{\pi} M$ and $N_\mu \xrightarrow{\mu} N$ is given by maps $\theta : M \to N$ and $\phi : M_\pi \to N_\mu$ such that $\theta \circ \pi = \mu \circ \phi$. Two bundles are isomorphic if there exist invertible bundle morphisms between them. Given a bundle $M_\pi \xrightarrow{\pi} M$ and a map $N \xrightarrow{f} M$, we get an induced bundle on N denoted by $f^*(\pi)$ via a pullback, that is, by defining the fiber of any point $x \in N$ to be the the fiber of $f(x)$ under the bundle π.

It is easy to check that the pullback actually gives a bundle over N. Let the total space of the pullback be denoted by $N_{f^*(\pi)}$. The pullback is a universal bundle on N in the sense that, if we have a bundle $N_\nu \xrightarrow{\nu} N$ on N and a smooth map $\phi : N_\nu \to M_\pi$ which is a bundle morphism, then, the map ϕ factors through $N_{f^*(\pi)}$.

Example 3.2. In this example, we try to explore the concept of a *universal* bundle in some sense. To start with, consider the Grassmannian $G_{n,k}$ of k-dimensional subspaces of $\mathbb{R}^n$. Then, we have the so-called tautological bundle over $G_{n,k}$, where, the fiber of any subspace is the set of point in that space. Formally, if we define

$$E_{n,k} := \{(x, L) : L \in G_{n,k}, x \in L\},$$

then, $E_{n,k}$ is a submanifold of $\mathbb{R}^n \times G_{n,k}$, and we have the projection along the second coordinate $\Theta : E_{n,k} \to G_{n,k}$. It is easy to see that Θ is actually a bundle, with fiber $\mathbb{R}^k$.

Next, consider any n-dimensional smooth manifold M and so, by Whitney's embedding theorem, we have an embedding $i : M \to R^{2n}$. For any $x \in \mathbb{R}^{2n}$, let r_x denote the translation by '$-a$' map on $\mathbb{R}_{2n}$. Then, for any $p \in M$, if we consider the map $r_{i(p)} \circ i : M \to \mathbb{R}^{2n}$, we get an induced

map on the tangent spaces $T_pM \to T_0\mathbb{R}^{2n}$, where $0 = r_{i(p)} \circ i$ denotes the origin in $\mathbb{R}^{2n}$. As i is an embedding, the map on tangent spaces is injective, and so, the image of T_pM under the above map is an n-dimensional subspace of $T_0\mathbb{R}^{2n}$. Let this subspace be V_p. So, we have a map from M to $G_{2n,n}$ given by

$$p \mapsto V_p,$$

which naturally also induces a morphism between the tangent bundle of M and the tautological bundle over $G_{2n,n}$ given by:

$$TM \to E_{2n,n}$$
$$(p, \xi) \mapsto (V_p, \xi),$$

where p is any point in M, and ξ is a tangent vector to M at p. After some thought, one notices that the above bundle morphism is induced by the map $M \to G_{2n,n}$. Thus, we conclude that given any n-dimensional manifold, its tangent bundle can be induced from the tautological bundle Θ of $G_{2n,n}$.

Definition 3.3. A vector bundle is a fiber bundle for which the fiber is a vector space V over $\mathbb{R}$, such that the trivialising diffeomorphisms are linear isomorphisms on each fiber.

Morphisms of vector bundles are usual bundle morphisms that are linear maps on the fibers. Similarly, for subbundles, fibers will be required to be subspaces of the original fibers and Whitney sums of vector bundles will involve the linear direct sum of the fibers as vector spaces. The added structure that a vector bundle π over a manifold M brings to a standard fiber bundle is the linearity. Having our fiber space as a linear space allows us to think of the space of sections $\Gamma(\pi)$ as a $C^\infty(M)$-module.

Remark 3.1. Example 3.2 can be generalised, where we replace the tangent bundle by any k-dimensional vector bundle E_π over M. By embedding E_π in $R^{2(n+k)}$ and identifying the tangent space of a linear space with itself, we get that E_π can be induced from the tautological bundle on $G_{2(n+k),k}$. This observation gives us the following corollary which will be useful in the next section.

Corollary 3.2. *Given a k-dimensional vector bundle E_π over an n-dimensional manifold M, the space of sections $\Gamma(\pi)$, as a $C^\infty(M)$-module, can be generated by at most $N_{n,k}$ generators, where $N_{n,k}$ depends only on n and k.*

Proof. By the above remark, the vector bundle can be induced from the tautological bundle over the Grassmannian $G_{2(n+k),k}$. Now, thinking of subspaces of $\mathbb{R}^{2(n+k)}$ as $k \times 2(n+k)$ matrices whose rows denote a basis for the subspace, the manifold $G_{2(n+k),k}$ can be covered by $\binom{2(n+k)}{k}$ open sets, depending on which $k \times k$ minors of the matrix are invertible. Restricted to each of these open sets, the tautological bundle Θ is trivial and can be spanned by k elements. Thus, $N(n,k) = k\binom{2(n+k)}{k}$ does the job. □

3.2 Serre-Swan Theorem

We have already talked about the space of sections $\Gamma(\pi)$ corresponding to a vector bundle $E_\pi \xrightarrow{\pi} M$. A moment of thought tells us that the correspondence Γ between vector bundles over M and $C^\infty(M)$-modules is actually functorial. For any bundle morphism $\phi : \pi \to \mu$ let $\Gamma(\phi) : \Gamma(\pi) \to \Gamma(\nu)$ be the induced map on the sections.

For any $p \in M$, let π_p denote its fiber space in E_π and recall that μ_p denotes the space of smooth functions on M that vanish at p.

Lemma 3.3. *Using the same notation as above,*

- *For any $y \in \pi_x$, there exists a section $s \in \Gamma(\pi)$ such that $s(x) = y$.*
- *Given $s \in \Gamma(\pi)$ such that $s(x) = 0$, there exist functions $f_1, f_2, \ldots, f_n \in \mu_z$ and sections $s_1, s_2, \ldots, s_n \in \Gamma(\pi)$, such that $s = \sum_{i=1}^{n} f_i s_i$.*

The proof is by observing that the result is obvious for trivial bundles, and then, generalising by using a partitions of unity argument. This lemma implies that we have a short exact sequence of the form:

$$0 \to \mu_x\Gamma(x) \to \Gamma(x) \to \pi_x \to 0, \tag{1}$$

where the first map is the inclusion, while the second one corresponds to evaluating a section at x.

Next, we talk about complements of subbundles of a given vector bundle. The following proposition, that motivates how projective modules come into the picture, is also of independent interest.

Proposition 3.4. *Given any vector bundle $E_\pi \xrightarrow{\pi} M$ and a subbundle μ of π, there exists a direct complement of μ in π, that is, there exists another subbundle ν of π such that $\pi = \mu \oplus \nu$.*

Proof. The argument hinges on the construction of a scalar product on the vector bundle π, where a scalar product is defined as an inner product on every fiber of the bundle that varies smoothly with the points on the manifold. One notices that scalar products are a generalization of the concept of a Riemannian metrics on the tangent bundle, and the existence of a scalar product for any vector bundle can be proved in exactly the same fashion as the existence of a Riemannian metric on a manifold: Observe that the result is obvious for trivial bundles, and then, glue together all the local scalar products using a partitions of unity argument.

Once we have a scalar product p on π, we can define a bundle on M such that for any point $x \in M$, the fiber is the complement of the subspace μ_x in π_x with respect to p. That these fibers will depend smoothly on x follows from the smoothness of p and the bundle μ. The vector bundle ν thus defined acts as the complement of μ in π. □

Now, we are ready to tackle the two main theorems of this report. The second one is the so-called *Serre-Swan theorem* that establishes an equivalence between the category of finite dimensional vector bundles over a manifold M and the category of finitely generated projective modules over $C^\infty(M)$.

Theorem 3.5. *Given vector bundles π and ν over a manifold M, we have an isomorphism as $\mathbb{R}$-vector spaces:*

$$Mor(\pi, \nu) \cong Mor_{C^\infty(M)}(\Gamma(\pi), \Gamma(\nu)).$$

Proof. The claim is that any $C^\infty(M)$-module map from $\Gamma(\pi)$ to $\Gamma(\nu)$ is uniquely of the form $\Gamma(\phi)$ for some morphism $\phi : \pi \to \nu$.

Let $f : \Gamma(\pi) \to \Gamma(\nu)$ be a module map. To construct a map $\phi : \pi \to \nu$ (i.e. $\phi : E_\pi \to R_\nu$), pick any $x \in E_\pi$. Then, by the surjectivity in Equation 1, there exists a section $s \in \Gamma(\pi)$ such that $s(\pi(x)) = x$. Define $\phi(x) = f(s)(\pi(x))$. This does not depend on the choice of the section s, because if $s_1(\pi(x)) = s_2(\pi(x))$, then, $s_1 - s_2 \in \mu_x\Gamma(\pi)$, and so, $f(s_1 - s_2) \in \mu_x\Gamma(\nu)$ implying that $f(s_1)(\pi(x)) = f(s_2)(\pi(x))$. Now, we claim that ϕ is actually a bundle morphism. This is observed by restricting to a common trivialising neighbourhood U of M with respect to both π and

ν. Then, the map $\phi|_{\pi^{-1}(U)} = (S_U)^{-1}(f)$, and the latter map can be expressed in terms of a matrix (fixing bases for the free $C^\infty(U)$-modules) with entries in $C^\infty(U)$, thus, proving the smoothness of ϕ locally, and completing the proof of the claim. Finally, tautologically, for any section $s \in \Gamma(\pi)$ and any $x \in M$, $(\Gamma(\phi)(s))(x) = \phi(s(x)) = f(s)(\pi(s(x))) = f(s)(x)$, and so, $\Gamma(\phi)(s) = f(s)$ for all $s \in \Gamma(\pi)$, i.e., $\Gamma(\phi) = f$.

In the end, uniqueness follows simply by observing that for morphims θ and ϕ, $\Gamma(\theta) = \Gamma(\phi)$ implies that $\theta \circ s = \phi \circ s$ for all sections $s \in \Gamma(\pi)$. Now, again by the right exactness in Equation 1, we conclude that $\pi = \phi$, thus, completing the proof of the theorem. □

Theorem 3.6. *Given a connected smooth manifold $M = |\mathcal{F}|$ (where, $\mathcal{F} = C^\infty(M)$, as usual), any $\mathcal{F}$-module P is isomorphic to the module of sections over a vector bundle over M if and only if P is projective and finitely generated as an $\mathcal{F}$-module.*

Proof. Suppose we start with a vector bundle $E_\pi \xrightarrow{\pi} M$. By Corollary 3.2, we have a finite set of section $s_1, s_2, \ldots, s_N$ that span $\Gamma(\pi)$. Let N be the free $C^\infty(M)$-module having the s_i's as basis elements. Then, N represents the space of sections for the trivial N-dimensional vector bundle ν on M. As there is an obvious map from N to $\Gamma(\pi)$, by the previous theorem, we get that it is induced by some morphism $\phi : \nu \to \pi$. Then, a routine regularity check gives us that $\ker(\phi)$ is a subbundle of ν. Hence, we get a direct sum decomposition $\nu = \ker(\phi) \oplus \eta$ for some bundle η. Then, checking at the level of fibers, we have $\eta_x = \pi_x$ for all $x \in M$, and thus, $\eta \cong \pi$' Hence, $N = \Gamma(\nu) = \Gamma(\ker(\phi)) \oplus \Gamma(\eta) = \Gamma(\ker(\phi)) \oplus \Gamma(\pi)$, which proves that $\Gamma(\pi)$ is a projective $C^\infty(M)$-module.

Next, let P be any projective $C^\infty(M)$-module. Then, there exists a free module N and a module Q such that $N = P \oplus Q$. Now, N can be assumed to be the module of sections of a trivial bundle η over M, i.e., $\Gamma(\eta) = N$. We attempt to find a subbundle of η that corresponds to the module P. For any $x \in M$, define $P_x := \{p(x) : p \in P \subset \Gamma(\eta)\}$, and similarly, we have Q_x. Then, by Lemma 3.3, we have that $\eta_x = P_x \oplus Q_x$ as $\mathbb{R}$-vector spaces. Next, $dim(P_x)$ is a locally constant function on a connected set, and thus, must be equal to a constant, say n for all $x \in M$.

We claim that using the P_x's as fibers, we get a subbundle of η that satisfies the required properties. To see that P_x varies smoothly with x, we fix an open neighbourhood U of x and choose a basis for P_x such that the corresponding elements in $\Gamma(\eta)$ are linearly independent in U, and thus, form a basis for all $y \in U$. Thus, the P_x's definitely form a bundle over M, which we will refer to as π. As sections of π locally resemble elements from the P_x's, the linear independence of P_x and Q_x implies that the sections actually belong to P, i.e., $\Gamma(\pi) = P$, thereby, completing the proof. □

Remark 3.2. We have glossed over some subtle technicalities in the proof involving the regularity of maps, but they all follow easily from the developed theory. We also note that the connectedness hypothesis is needed for only one of the implications of the theorem, i.e., the module of sections is projective and finitely generated for arbitrary smooth manifolds.

An easy consequence of the above theorem is the following, geometrically non-intuitive corollary which follows from the algebraic definition of projectivity and functoriality.

Corollary 3.7. *Given any vector bundle π, there exists another bundle μ such that $\pi \oplus \mu$ is the trivial bundle.*

4 Algebraic geometry

In this section, we see an analogue of the Serre-Swan theorem in the context of vector bundles over affine schemes. While this is not directly related to what we have been discussing till now, it goes to show how similar concepts arise in various realms of mathematics and often behave in a coherent fashion, giving an indication towards the existence of deeper connections that may not be entirely conspicuous.

4.1 Correspondence between vector bundles and locally free sheaves

We set some preliminary notation. Fix a field k. Throughout the section, all fiber products will be taken over $Spec(k)$. We denote by $\mathbb{A}^n$ the affine scheme $Spec(k[x_1, x_2, \ldots, x_n])$. For any scheme U over $Spec(k)$, we use $\mathbb{A}^n_U$ to denote the fiber product $U \times \mathbb{A}^n$.

Definition 4.1. Given a scheme Y (not necessarily affine), a vector bundle of rank n over Y is a scheme X with a map $\phi : X \to Y$ such that we can cover Y using open sets $\{U_i\}$ such that for all i, $\phi^{-1}(U_i) \cong \mathbb{A}^n_{U_i}$, and for any affine open sets $Spec(A) \subset U_i \cap U_j$, we get that the transition map $\phi_i \circ \phi_j^{-1}$ induces on an automorphism $\mathbb{A}^n_V = Spec(A[x_1, x_2, \ldots, x_n])$ induced by a linear change of coordinates on the ring $A[x_1, x_2, \ldots, x_n]$.

The above definition is immediately seen to be similar to the definition of vector bundles over smooth manifolds, in the sense that the base space can be covered by open sets such that the fiber of each open set U has the form of a trivial bundle ($U \times \mathbb{A}^n$ in our case), and the transition maps are *linear* in some sense. We call two vector bundles on a given scheme X isomorphic if there is an isomorphism between the underlying schemes that commutes with the projection maps and such that the open coverings of X that are used for defining the two vector bundle structures are compatible with each other.

Now, given a scheme $(X, \mathcal{O}_X)$, a sheaf of $\mathcal{O}_X$-modules is free if it is isomorphic to the direct sum of copies of $\mathcal{O}_X$. $\mathcal{F}$ is a locally free sheaf of rank n if we have an open covering $\{U_i\}$ of X such that $\mathcal{F}|_{U_i}$ is a free $\mathcal{O}_X|_{U_i}$-module of rank n. Our goal will be to connect such locally free sheaves of rank n to vector bundles of rank n. This is done through the following lemmas.

Lemma 4.1. *Given $\mathscr{E}$ a locally free sheaf of rank n on a scheme Y, we can naturally associate with it a vector bundle of rank n.*

Proof. Consider $S(\mathscr{E})$, the symmetric algebra over $\mathscr{E}$, and let $X = Spec(S(\mathscr{E}))$. Then, covering Y by affine subschemes if needed, so that the restricted sheaf $\mathscr{E}$ is free for those subschemes, we get a morphism $f : X \to Y$ by gluing together pieces from all the subschemes. We claim that X can be given a vector bundle structure over Y.

To see this, let $U_i \subset Y$ be such that $\mathscr{E}(U_i)$ is a free $\mathcal{O}_Y(U_i)$-module, and so, we can identify $S(\mathscr{E}(U_i))$ with $\mathcal{O}_Y(U_i)[x_1, x_2, \ldots, x_n]$, after the choice of a suitable basis. This gives an isomorphism ϕ_i from $f^{-1}(U_i)$ to $\mathbb{A}^n_{U_i}$, which provides a covering of Y by open sets such that the restriction of f to these sets corresponds to a trivial bundle. Next, if $V = Spec(B) \subset U_i \cap U_j$, then the automorphism induced on $Spec(B[x_1, x_2, \ldots, x_n])$ is the appropriate restriction of $\phi_j \circ \phi_i^{-1}$ which, by construction, is induced by a linear change of coordinates. Thus, we conclude that the map f makes X a vector bundle over Y of rank n. □

Given a morphism $f : X \to Y$, for any open subset $U \subset Y$, a section of f over U is a morphism $g : U \to X$ such that $f \circ g = id_U$. As morphisms of schemes can be glued together, the assignment to U of the set of all sections of f over U gives us a sheaf on Y, which we refer to as the sheaf of sections of f and denote by $\mathscr{S}(X/Y)$.

Lemma 4.2. *Given X, a vector bundles of rank n over Y, the sheaf of sections $\mathscr{S}(X/Y)$ has a natural $\mathcal{O}_Y$-module structure, that is locally free of rank n.*

Proof. Without loss of generality, we can assume that Y is an affine scheme, that is, $Y = Spec(A)$ for some ring A. This is because we can cover Y with with affine subschemes, and glue together the respective sheaves of modules to get a sheaf for all of Y. Choosing a smaller open subset of Y if needed, we can assume that X is a trivial bundle of rank n over Y, i.e., $X = \mathbb{A}^n_Y$.

Now, any morphism between affine schemes is induced by a ring homomorphism in the opposite direction, and so, any global section of $f : \mathbb{A}^n_Y \to Y$ comes from a ring homomorphism $\phi : A[x_1, x_2, \ldots, x_n] \to A$. As such a homomorphism is determined by an n-tuple of elements of A, we can identify $\Gamma(Y, \mathscr{S}(X/Y))$ with A^n, which gives us a natural A-module structure on $\mathscr{S}(X/Y)$. Also, the identification implies that $\mathscr{S}(X/Y) \cong \mathcal{O}_Y^n$, and thus, is free of rank n. Hence, gluing up all the open subsets together, we get a locally free sheaf of rank n, as needed. □

We claim that the above correspondences are actually (almost) the inverses of each other. To see this, suppose for a given sheaf $\mathscr{E}$ of $\mathcal{O}_X$-modules, we denote by $\check{\mathscr{E}}$ the dual sheaf of $\mathscr{E}$, that is, the sheaf $Hom_{\mathcal{O}_X}(\mathscr{E}, \mathcal{O}_X)$. Then, if we start with a locally free sheaf $\mathcal{F}$ of rank n, apply lemma 4.2 to get a vector bundle, and then, use lemma 4.3 to get back to a sheaf, we see that the construction of sections in lemma 4.3 means that the sheaf that we obtain is $\check{\mathcal{F}}$. Now, as $\check{\check{\mathcal{F}}} = \mathcal{F}$, we conclude that the above correspondence can be reversed, and so, we have the following result:

Proposition 4.3. *There is a one to one correspondence between isomorphism classes of locally free sheaves of rank n and isomorphism classes of vector bundles of rank n over a given scheme.*

Thus, there is a tendency in algebraic geometry to talk about vector bundles and locally free sheaves interchangeably, which is often quite helpful in gaining useful insights, and using geometric intuition to think about abstract algebraic objects.

4.2 Serre's Theorem

Given an affine scheme $X = Spec(A)$ with structure sheaf $\mathcal{O}_X$ and an A-module M, we can get a sheaf of $\mathcal{O}_X$-modules corresponding to M, which we will denote by $\tilde{M}$. For an arbitrary scheme $(X, \mathcal{O}_X)$, a sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules is called quasi-coehernt if X can be covered by open affine subsets $U_i = Spec(A_i)$, such that $\mathcal{F}|_{U_i} = \tilde{M}_i$ for some A_i-module M_i. The sheaf is called coherent if it is quasi-coherent and each of the above M_i's can be taken to be a finitely generated A_i-module. Before moving on, we state the following lemma, which is often used as an alternative definition for coherent and quasi-coherent sheaves:

Lemma 4.4. *Given a scheme X, a sheaf of $\mathcal{O}_X$-modules $\mathcal{F}$ is quasi-coherent if and only if for all affine open subsets $U = Spec(A)$ of X, $\mathcal{F}_U = \tilde{M}$ for some A-module M. If X is Noetherian (i.e., X can be covered by finitely many affine open subsets), then $\mathcal{F}$ is coherent if and only if the above is true along with M being finitely generated as an A-module.*

The main result of this section is the following theorem by Serre, which was first published by him in 1955 (see [3]), historically preceding the result we saw in the smooth manifold setting. It gives a condition dictating when coherent sheaves over an affine scheme give finite dimensional vector bundles under the correspondence discussed in the previous section. As vector bundles of rank n correspond to locally free sheaves of rank n, we wish to know when coherent sheaves are locally free. By the lemma above, any coherent sheaf over $X = Spec(A)$ must be of the form $\tilde{M}$ for some finitely generated A-module M. Thus, the coherent sheaf is locally free if and only if M is a locally free A-module which, by an exercise in commutative algebra, happens if and only if M is a projective A-module. This gives us all the pieces to conclude with the following theorem:

Theorem 4.5 (Serre)**.** *Given an affine scheme $(X = Spec(A), \mathcal{O}_X)$ and a coherent sheaf $\mathcal{F}$ of $\mathcal{O}_X$-modules, $\mathcal{F}$ corresponds to a finite dimensional vector bundle over X if and only if the space of global sections $\Gamma(\mathcal{F}, X)$ is a projective A-module.*

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